2 Clinical detection: 'information gathering'

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2.1 Introduction

The oral healthcare team (dentist, nurse, hygienist/therapist/oral health educator, laboratory technician, receptionist, practice manager), led by the principal dental practitioner, should all be involved in the decision-making processes and dental management of the patient, as part of the minimum intervention philosophy of oral/dental healthcare (see Figure 1.1 in Chapter 1). This care rationale is patient-centred, engaging with the patient to encourage them to take responsibility for their own oral health. The role of the oral healthcare team is to provide advice and guidance to help the patient to maintain oral health, as well as providing operative treatment to repair damaged hard and soft tissues. Sometimes the dentist will refer difficult cases to a specialist dentist for their opinion as to what the diagnosis and care plan should be.

To manage patients successfully, there are five stages that must be followed (see Figure 2.1):

1. Detecting clinical problems and their aetiology (see Chapter 2):
   - This involves detective work to help to gather clinically relevant and useful information, primarily using the skills of verbal history taking, oral examination, and relevant special investigations.
2. Diagnosis and risk assessment (see Chapter 3):
   - The art of interpretation of signs and symptoms/results from investigations to conclude with identifying the cause of the problem and the potential the individual patient has of developing further disease in the future or responding to treatment. Both aspects are critical to planning the overall care of the patient.
3. Prognosis (see Chapter 3):
   - The art of forecasting the course of a disease or problem, whether treated or not.

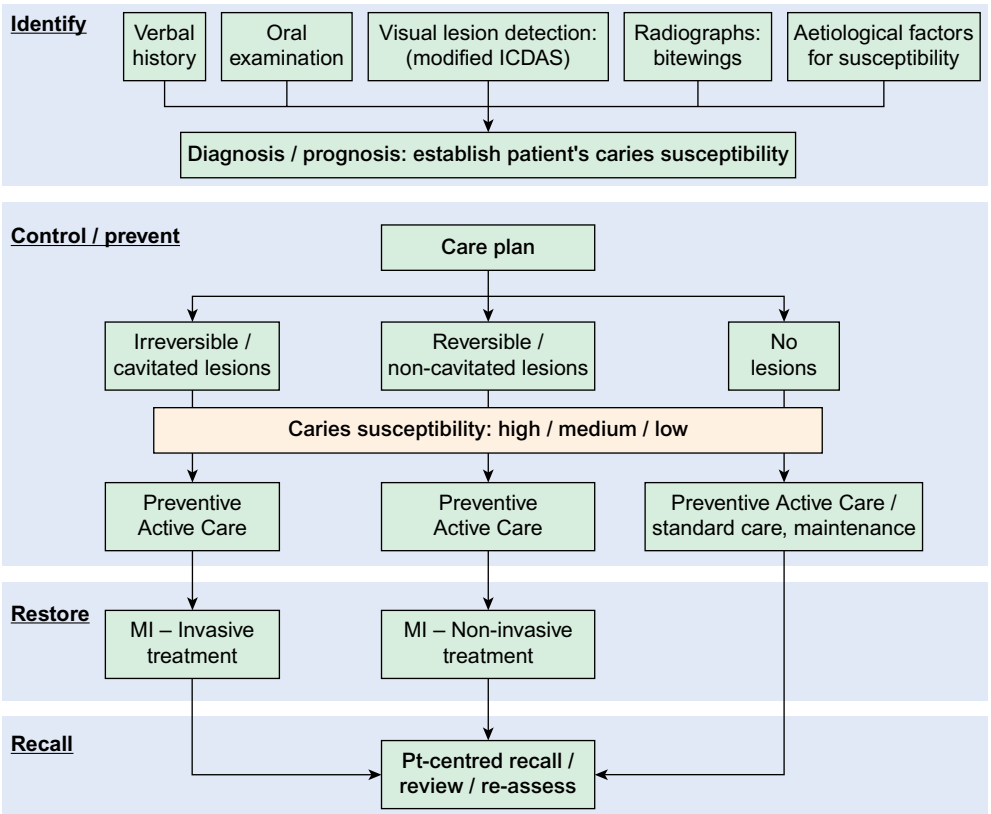


Figure 2.1 Minimum intervention patient care flowchart showing how the patient management stages of *identify* (see Chapter 2), *control and prevent* (see Chapter 4), *restore* (see Chapters 5, 6, 7, and 8), and *review/monitor/recall* (see Chapter 9) link to one another (see also Figure 1.1 in Chapter 1). Note how the patient-centred care plan focuses on caries risk/susceptibility. The medium risk grade has been shaded to signify that at a practical patient contact level, most can be managed as either high or low risk. The key factor is that the patient's risk and susceptibility are monitored over time. Tooth wear and other conditions will be considered later in this chapter. (Reproduced from A Banerjee and S Domejean. *Primary Dental Journal, 2, 3,* pp. 30–37. © 2013, with permission from the Royal College of Surgeons.)

4. Formulation of an individualized patient care plan (see Chapters 3 and 4):
   - This must be underpinned by the non-invasive control of disease and lesion prevention, following the principles of minimum intervention oral care.
   - The care plan will also include itemized, costed, minimally invasive operative treatments when required.
5. Recall/re-assess/review (see Chapter 9):
   - Reviewing the outcomes of any care provided, re-assessing the patient's response to evaluate whether knowledge/behavioural adaptations and adherence have helped to control and/or prevent disease reoccurrence, and developing adaptive recall strategies/intervals that are patient-centred, rather than generic and guideline-driven.

2.2 Detection/identification: 'information gathering'

This aspect of patient management involves gathering the relevant information based on which sound clinical judgements can be made as to the best course of treatment for the individual patient. Clinical detection works on two levels:

- detecting the immediate clinical problem (i.e. trauma or the physical manifestation of the disease process, such as carious lesions, tooth-wear (TW) lesions)
- detecting and understanding the complex interplay of aetiological factors that have *caused* the problem for that particular patient over time.

It is critical that detailed notes are made for all stages of the patient management pathway, documented at every visit. This can be done using electronic patient recording systems or handwritten notes. Each patient will have their own notes with their name, date of birth, address, contact details, and occupation listed. It is essential to check that the correct notes have been called up for the attending patient, and for each visit the date and time of appointment must be noted initially.

2.3 Taking a verbal history

The art of verbal history taking (anamnesis) is the first and most important way of obtaining information about any problems the patient may present with, and also their aetiology. The answers to a series of carefully planned questions will help to shed light on the nature and severity of the problem(s) presented. A generic structured 'interview' with the clinical relevance of each question is presented in Table 2.1. Of course this is not the only format of questioning that can be used, and with experience it should be adjusted to the specific needs and conditions of the patient, and modified to assess aetiological factors of the presenting conditions (see later).

Table 2.1 Structured verbal history ordered to ask relevant questions to help unearth the clues for a diagnosis of the presenting problem and the formulation of an individualized care plan

Structuring the verbal history	Comments
Patient complains of . . . (C/O)	Document, in the patient's own words, the presenting symptom(s)/problem(s)
History of presenting complaint(s) (HPC)	*Commencement*: when did it/they start? *Location*: ask the patient to describe or ask them to point/outline the area with one finger *Type*: description of symptoms. Avoid putting words in the patient's mouth *Incidence*: how long ago did the episodes start? *Duration*: for how long do they persist? Frequency? Are they getting better, staying the same, or deteriorating? *Initiating/relieving factors*: does anything make the symptoms worse or better? **Answers to the above will often provide the clues to help direct the clinician to the correct diagnosis**
Past dental history (PDH)	What previous dental treatment has the patient experienced (orthodontics/extractions/periodontal treatment/fillings, etc.)? How regularly do they visit the dentist? What previous preventive advice have they received/do they follow? **Answers to the above will help to create a picture of the attitude/motivation of the patient towards dentistry and their own oral health, without asking them directly about these issues**

Medical history (MH)	Most oral healthcare teams use a formatted checklist, which should include information about: • cardiac problems/disease/rheumatic fever/blood pressure • respiratory disease/asthma/shortness of breath • diabetes/epilepsy/jaundice /hepatitis history • current/recent past medications • allergies • bleeding/haemorrhage/clotting defects • other illnesses/operations/hospital admissions • pregnancy • HIV/AIDS/communicable disease risk **Information relevant regarding drug interactions with local anaesthetic, and allergies to latex/resins/dental materials. Bleeding problems/anticoagulant therapies relevant for periodontal surgery, root planing, and extractions. Check medication in the *Dental Formulary* or equivalent. Medications that cause dry mouth, gingival overgrowth, or vomiting (emesis) should be noted**
Social history (SH)	Occupation/family members/availability for appointments **Relevant when considering appointment logistics to deliver the care plan for the individual**
Habits	Oral hygiene (OH—procedures, frequency), use of fluoride, diet (amount/frequency of sugar intake, balanced diet, erosive potential for tooth wear), smoking, alcohol intake, parafunctional habits (bruxism (teeth grinding), clenching) **Relevant when planning care and giving preventive advice. What the patient says might be checked/verified during the oral examination to follow. Helps to ascertain the possible aetiology of the problem**

2.4 Physical examination

Once the verbal history has been taken, a physical examination follows, which is divided into a general and oral examination.

2.4.1 General examination

This will commence as the clinician interviews the patient. The demeanour of the individual, facial/ocular asymmetries, facial nerve palsies, pallor, tremors, or mental/physical disabilities can be noted and followed up if relevant to the dental care required.

2.4.2 Oral examination

See Table 2.2. An error commonly made by inexperienced clinicians is to dive into an oral examination and direct all attention immediately to the teeth. It is better to have a system that encompasses all aspects of oral health, including the mucosal soft tissues and periodontal tissues, as well as teeth and prostheses. When examining the oral mucosae, learn a system which ensures that all aspects of the oral cavity are included, i.e. anterior to posterior (lips to tonsils), posterior to anterior, or clockwise/anticlockwise around the oral cavity. Once learned and practised, this is never forgotten, even during the stress of clinical examinations! Ensure that adequate time has been set aside for the examination, along with suitable dental instrumentation and lighting (see Chapter 5). Clinical digital photography can often assist in the documentation of any notable findings during the examination, but appropriate informed and written consent must be obtained prior to any images being captured and stored securely. Data to be recorded along with the image(s) must include patient details, the date of capture of the image(s), the clinical findings, and, where appropriate, a scale to help the viewer to assess the size of any lesion imaged. The camera settings and ambient lighting should also be noted in case comparative images are taken at a later date, to enable standardization between them. The assisting member of staff should be trained to document accurately and faithfully all of the information gathered during this part of the examination. However, the practising clinician still has ultimate responsibility for checking any records taken during the examination.

The examination of the oral cavity must also include an assessment of the quantity and quality of the patient's saliva. As has been previously mentioned, saliva plays a critical role in the control of dental disease risk, especially caries and erosion (see Section 2.5.4).

2.4.3 Dental charting

When examining the dentition, there are certain features that the dentist must relay to their assistant to record accurately in a dental chart (see Figure 2.2). This chart is interpreted as though the dentist is looking

Table 2.2 Steps involved in a comprehensive clinical oral examination. Note that a full examination should not be rushed, and careful recording of findings is essential both clinically and dento-legally

Examination site		Comments
Extra-oral		Facial swellings/asymmetries, lips—form and seal, facial and neck lymph nodes, TMJ—crepitus, clicking (uni-bilateral), mandibular movement (opening gape, deviations, lateral/protrusive excursions) **Operator stands behind the patient for the TMJ/LN exam—remember to notify the patient prior to starting!**
Intra-oral	Mucosae	All internal buccal, labial, alveolar mucosae, including vermilion border of lips, tongue (dorsum, lateral borders, and ventral surfaces), retromolar areas, hard and soft palate, and floor of mouth. Palpation of the pterygoid muscles **Checking for white or erythematous patches/plaques indicative of trauma, lichenoid reactions, or neoplastic change—referral to an oral medicine specialist might be appropriate. Tenderness in the pterygoid muscles—sign of TMD? Dry mucosae and frothy saliva are indicative of dry mouth (see Section 2.5.4)**
	Periodontium	Marginal gingivae (colour, contour, consistency), gingivitis (BPE score), recession, loss of attachment, probing depths, mobility, presence of supra-/subgingival calculus, plaque indices **Relevant to assess periodontal status, which will affect the overall restorative status of the mouth and of the individual tooth**
	Teeth	Missing teeth, mobility, restoration status, caries, tooth wear (site, enamel ± dentine), malpositioning (tilting, rotation, overeruption/submerged)
	Prostheses	Crowns, removable dentures, fixed bridges, implant-retained crown and bridge work, orthodontic appliances (fixed and removable) **Relevant regarding oral hygiene procedures, plaque-retentive margins, aesthetics, and status of abutment teeth**
	Occlusion	Angle's classification (Class I, II, and III), incisor relationship (Class I, II division 1/2, and Class III). Intercuspal position (ICP), retruded contact position (RCP), protrusive, retrusive, lateral excursive movements—working and non-working side contacts. Skeletal discrepancies **Relevant to the restoration of individual/groups of teeth to conform to the existing occlusal harmony or to assess changes in a reorganized approach in more complex care plans (see Chapter 5)**
	Saliva (see Table 2.5)	*Quality:* normal, frothy, viscous. *Quantity:* normal, reduced output, buccal mucosae/tongue sticks to mirror head, limited/no pooling in floor of mouth, shiny mucosae lobulated/fissured tongue, altered gingival architecture, glassy appearance of oral mucosae, cervical caries, food debris present **Relevant regarding the protective effects of saliva for teeth and the oral mucosae. Low saliva output/poor quality increases patient risk of caries and oral infections**

TMJ, temporomandibular joint; LN, lymph node; TMD, temporomandibular dysfunction; BPE, Basic Periodontal Examination.

at the patient face to face, and the notation is communicated from the patient's perspective.

1. Which tooth, side, position (e.g. 'upper left 6' or 'upper left first permanent molar', rotated/tilted/space-closed).
2. Presence of existing restoration(s)—location (mesio-occlusal, buccal, cervical, etc.) and type of material (amalgam, tooth-coloured, gold, ceramic, etc.).
3. Status of existing restorations—sound, or deficient margins/fractured/missing.
4. Presence of carious lesion(s)—location, mICDAS classification (see Table 2.3), lesion activity.
5. Presence of tooth wear—location (buccal, occlusal, incisal) and extent (in enamel or dentine, exposing pulp, BEWE score) (see Table 2.7).
6. Presence of other abnormalities (e.g. fluorosis, hypoplastic enamel, cracks).

With practice, this can be done with a series of abbreviated responses, and good teamwork with the nurse/assistant can expedite

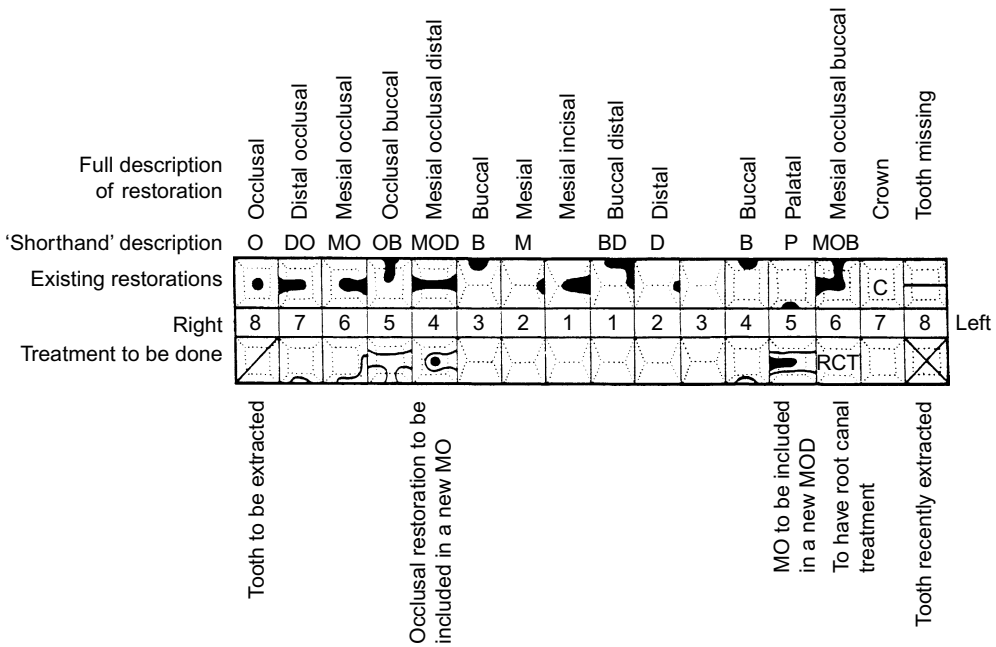


Figure 2.2 Conventions for recording restorations, lesions/teeth requiring restoration, teeth to be extracted, and other conditions on a dental chart.

Q2.1: Can you spot an error in the charting in Figure 2.2?

this process. Of course, if points 2–6 in this list do not apply after careful examination, a summary of 'sound' will usually suffice and a dot is placed on the relevant tooth in the chart. It is important to check all five surfaces of each tooth either with direct vision or using the dental mirror and good lighting of a clean tooth. It might be necessary in some patients, before the visual examination (but after checking any relevant medical history), to clean the dentition with a periodontal hand/ultrasonic scaler in order to remove supragingival plaque and calculus deposits that might be obscuring the clear view of the tooth surfaces beneath.

2.4.4 Tooth notation

In the UK, the most commonly used notations are the Palmer system and a two-letter coding system. Both are shown in Figure 2.3, dividing the mouth into four quadrants with the teeth in each numbered from 1 to 8 from central incisors to third molars.

A drawback of the Palmer notation is that this can be difficult to input into electronic notes (although current software packages are accommodating this system).

Two numerical notation systems exist—the FDI (Federation Dentaire Internationale, favoured in mainland Europe; see Figure 2.4) and the Universal system (favoured in the USA; see Figure 2.5). Users tend to credit these systems for crossing language boundaries, but confusion can easily occur with these solely numerical notations—for example, FDI 16 (see Figure 2.4) represents UR6 (upper right first permanent molar, 6|), whereas Universal 16 (see Figure 2.5) represents UL8 (upper left third permanent molar, |8).

UR	8	7	6	5	4	3	2	1	1	2	3	4	5	6	7	8 UL
LR	8	7	6	5	4	3	2	1	1	2	3	4	5	6	7	8 LL

Figure 2.3 The Palmer and two-letter coding systems (UR, upper right; UL, upper left; LR, lower right; LL, lower left) communicated as though the dentist is looking directly at the patient. In both systems the deciduous dentition is labelled a–e from the midline.

UR	18	17	16	15	14	13	12	11	21	22	23	24	25	26	27	28	UL
LR	48	47	46	45	44	43	42	41	31	32	33	34	35	36	37	38	LL

Figure 2.4 The FDI tooth notation system with the teeth in each quadrant prefixed by a number from 1 to 4.

UR	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	UL
LR	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	LL

Figure 2.5 The Universal tooth notation system, with each individual tooth being allocated a number from 1 to 32 in the permanent dentition.

Table 2.3 A modified ICDAS (mICDAS) carious lesion scoring system (0–4), linking the clinical appearance with the equivalent underlying lesion histology. Images show teeth sectioned longitudinally through occlusal lesions, as representative clinical examples of each mICDAS score. This clinical scoring system is useful for inclusion in the patient's notes, for monitoring, and for dento-legal purposes.

0		Clinical appearance: No or slight change in enamel translucency after prolonged air drying (> 5 s) **Underlying histology: No enamel demineralization or a narrow surface zone of opacity**
1		Clinical appearance: Opacity or discoloration from the enamel (white spot lesion) hardly visible on a wet surface, but distinctly visible after air drying. No cavitation on occlusal/smooth surfaces **Underlying histology: Enamel demineralization limited to outer 50%**
2		Clinical appearance: Enamel opacity (white spot lesion) or greyish discoloration distinctly visible without the need for air drying. No clinical cavitation detectable **Underlying histology: Demineralization involving inner 50% of enamel through to the outer third of dentine**
3		Clinical appearance: Localized enamel breakdown in opaque or discolored enamel, ± greyish discoloration/shadowing from underlying dentine **Underlying histology: Demineralization involving the middle to inner third of dentine**
4		Clinical appearance: Gross cavitation in opaque or discoloured enamel exposing the underlying stained dentine **Underlying histology: Demineralization involving the inner third of dentine towards pulp**

2.5 Caries detection

Visual detection of carious lesions in enamel and dentine relies on the following operator-controlled factors:

- Using 'sharp' eyes and the recommended use of magnification in the form of dental loupes (see Chapter 5).
- Using good illumination from the overhead dental chair light or a more focused light from an LED headlight, usually coupled with the use of magnification loupes.
- Having clean tooth surfaces to examine both wet and dry (using a 3-1 air/water syringe; see Figure 2.6(a) and (b)). If surface debris (plaque/calculus) is present, this may have to be removed prior to any dental examination taking place (see Figure 2.6 (a) and (b) and Figure 1.2 (b) and (c)).
- Using rounded/ball-ended dental explorers—the use of sharp dental probes is contraindicated for carious lesion detection as they can, with injudicious use, cause cavitation in a previously non-cavitated lesion (see Figure 2.7).
- Separating teeth may be beneficial to assess the proximal surfaces of adjacent teeth visually, ideally with magnification. Wedges or orthodontic separators can be placed interproximally for a few minutes prior to the examination of the field, to gently displace the gingiva apically and part the teeth slightly through movement within their periodontal ligament spaces. In this way, incipient or cavitated lesions may become more evident.
- Sufficient time allocated for the examination of all tooth surfaces as well as the soft tissues and periodontium.

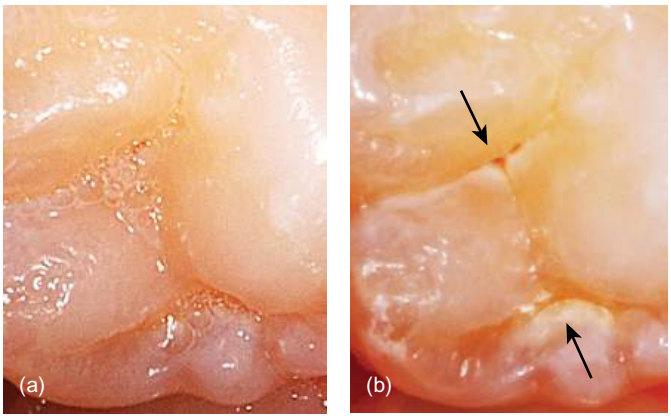


Figure 2.6 **(a)** A maxillary molar that is apparently caries-free with saliva obscuring the occlusal fissures. **(b)** When the saliva is removed with the 3-1 air/water syringe, an incipient white spot lesion becomes evident (mICDAS 1). (Courtesy of L. Mackenzie.)

Q2.2: What hand instrument (in Fig 2.7) should be used instead and for what purpose?

2.5.1 Caries detection indices

The histopathology and clinical signs of the carious lesion have been described in Chapter 1. The clinical manifestation of the caries process is the progressive lesion it creates within the dental hard tissues. A dentist must be able to link the visual clinical appearance of the lesion with the underlying histological damage that has occurred in the tooth (and possibly lesion activity status), at a particular moment in time. In this way the dentist can then diagnose problems, and decide how to manage the lesion/disease process in that individual patient through the development of a suitable care plan. In an attempt to do this, as well as for reviewing the relative success or failure of previous treatment at recall consultations and to help with dento-legal documentation, several visual indices have been described over the years. In 2004, the ICDAS (International Caries Detection and Assessment System) Foundation was convened to produce an evidence-based clinical caries assessment system to be used primarily for epidemiological and research studies, as well as in general dental practice (see QR code image 2.1).

www.icdas.org

QR code image 2.1 Scan this code with your mobile device to view the International Caries Detection and Assessment System.

A simpler, modified version is presented in Table 2.3 with clinical examples (see Figures 2.6, 2.8, 2.9, and 2.10), which permits the dentist to examine clinically the tooth surface and appreciate the underlying damage that has been caused. Then, depending on the individual's caries risk assessment, the most relevant treatment option can be chosen. This index requires that the points listed and discussed previously are implemented judiciously, and allows an objective numerical record to be made in the dental chart to permit longitudinal, dento-legally appropriate assessment of the particular lesion over time.

Alternative indices exist for classifying carious lesions. The 'caries iceberg', developed in collaboration with cariologists and epidemiologists, is an index used by many experts to study the incidence and

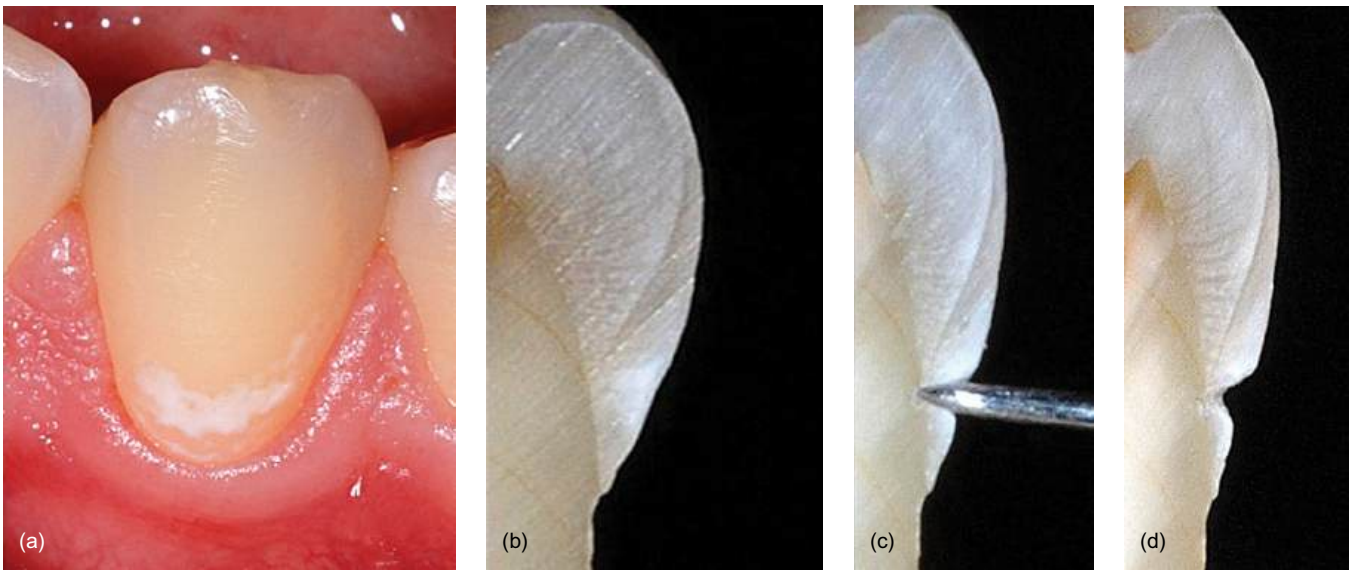


Figure 2.7 **(a)** A smooth surface incipient white spot lesion (WSL) associated with the cervical aspect of a mandibular canine. **(b)** A cross-section through an equivalent buccal cervical WSL showing the classical 'inverted cone' shape of the non-cavitated enamel lesion (see Chapter 1, Section 1.2.6). **(c)** This lesion has been probed using a sharp dental explorer resulting in **(d)** operator-induced (iatrogenic) cavitation of the lesion that should have been managed non-operatively. The use of a sharp dental probe to elicit 'sticky fissures/surfaces' is contraindicated for caries detection. (Courtesy of L. Mackenzie.)



Figure 2.8 The brown, non-cavitated fissures with underlying grey discoloration on the occlusal surface of the UL7 are caused by demineralized, discoloured dentine shining through intact, wet enamel (a closed lesion, visible in dentine on the bitewing radiograph (right, yellow arrow)—mICDAS 2).

Q2.3: What treatment may be required for the tooth shown in Figure 2.8?

prevalence of the disease in different populations. This permits the development of strategies to manage caries at a population level as well as at a patient level. The 'iceberg' can be seen in Figure 2.11—clinically detectable lesions are divided into four groups, D1–D4, depending on the depth of tissue invasion and the degree of cavitation. Note that careful selection of caries detection thresholds is required, otherwise this index may lead to threshold bias when interpreting data for caries prevalence in a population. For example, if D3 detection threshold (dentine lesions, cavitated or not, i.e. open or closed) is selected to establish the presence/absence of caries, patients with early-stage lesions within enamel, cavitated or not, will not be included in the final data, thus incorrectly reducing the caries prevalence for that particular population.

2.5.2 Susceptible surfaces

Carious lesions occur on tooth surfaces that have accumulated plaque biofilm which has been allowed to stagnate for a prolonged period of time. During the clinical examination, it would be prudent to examine these areas with particular care, for early signs of lesions:

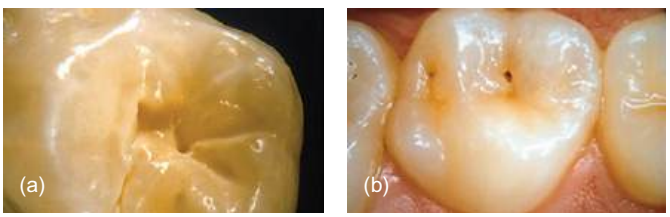


Figure 2.9 **(a), (b)** A small cavity is just detectable within the whitish opacity/brown discoloration on the occlusal surfaces of these molars, appearing as a widened fissure and a cavitated pit, respectively. They can easily be missed clinically unless the surfaces are clean and vision aided by the use of magnification. Histologically, both lesions were well into dentine, visible on radiographs—mICDAS 3.



Figure 2.10 Grossly cavitated buccal carious lesions on LR3 and LR4, with quantities of stagnant dense plaque deposited on the exposed dentine—mICDAS 4.

- The depths of pits and fissures on posterior occlusal surfaces (which cannot be cleaned effectively with a toothbrush). These areas on newly erupting, partially erupted, or submerged molars are particularly susceptible to carious attack.
- Proximal surfaces (mesial and distal) *cervical* to contact points of adjacent teeth (where patients may not floss regularly, or at all). The surfaces of particularly imbricated (crowded) teeth can be more susceptible due to the lack of access to oral hygiene aids.
- Smooth surfaces adjacent to the gingival margin (again where patients often miss with their toothbrush).
- The ledged/overhanging/deficient margins of existing restorations (a plaque trap often inaccessible to a toothbrush/floss).

Of course, these are not the only sites on which carious lesions occur. The site and distribution of lesions in a patient's mouth might give an indication of underlying aetiological factors. For example, patients with xerostomia (dry mouth) due to salivary gland disease or damage (e.g. from radiotherapy when treating head and neck cancers) can develop lesions on the incisal surfaces of anterior teeth and lesions that circumvent the neck of the crown at the gingival margin. Patients with eating disorders may have lesions on the lingual-cervical aspect of the mandibular teeth (see Figure 2.12).

2.5.3 Special investigations

To aid the process of clinical information gathering, other investigations may be requested, the results of which must be interpreted clinically to help to verify or make the final diagnosis of the problem and possibly its aetiology. Investigations have a cost implication for the patient or the health service provider and, depending on the test, may be invasive or even potentially harmful in nature. It is imperative that information

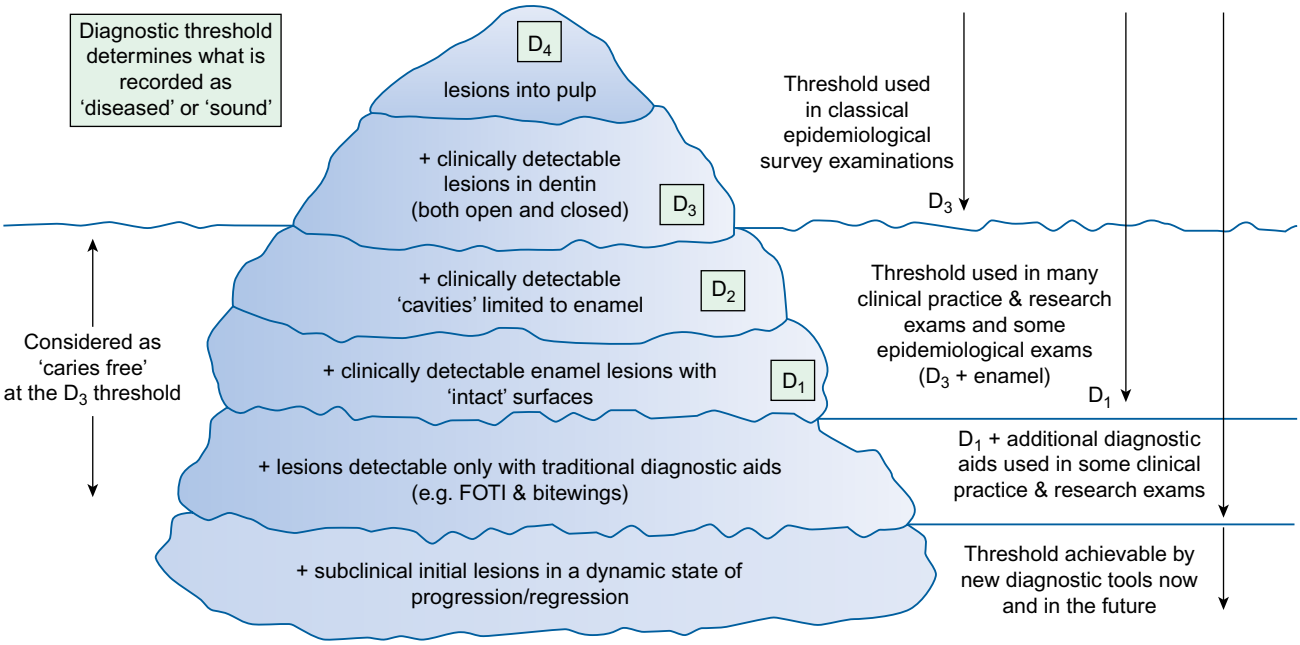


Figure 2.11 The 'caries iceberg' classifying clinically detectable lesions with regard to their level of tissue invasion and degree of cavitation. (Reproduced from NB Pitts 'Diagnostic tools and measurements – impact on appropriate care.' *Community Dentistry and Oral Epidemiology,* 25, pp. 24–35. © 1997, with permission from John Wiley & Sons Ltd.)

gathered using multiple special investigations is not interpreted individually, but in conjunction with all of the other methods described in this chapter to help to formulate the diagnosis for the patient. All investigations are prone to false-positive and false-negative outcomes. Analysing the relative proportions of these outcomes, statistical measures of *sensitivity* (the measure of how effective an investigation is at detecting true disease), *specificity* (the measure of how effective an investigation is at detecting true health), and the *positive predictive value* (*ppv*—that is, *true positives divided by the sum of true and false positives)* can be developed, which are useful in assessing the investigation's value in providing a clinically relevant diagnostic yield. For caries detection, these include intra-oral radiographs, pulp vitality (sensibility) and percussion tests.

Radiographs

Horizontal bitewing radiographs should be used to aid lesion detection on proximal surfaces of posterior teeth, especially when adjacent teeth are present and direct vision is not possible. A film holder and beam-aiming device should be used routinely in order to obtain the optimal angulation of the beam perpendicular to the contact points and to allow reproducibility of films when monitoring lesions over a period of time (see Figure 2.13).



Figure 2.12 A patient with an eating disorder, anorexia nervosa, with carious cavitated lesions on the lingual aspects of LL124 (mICDAS 2,3).

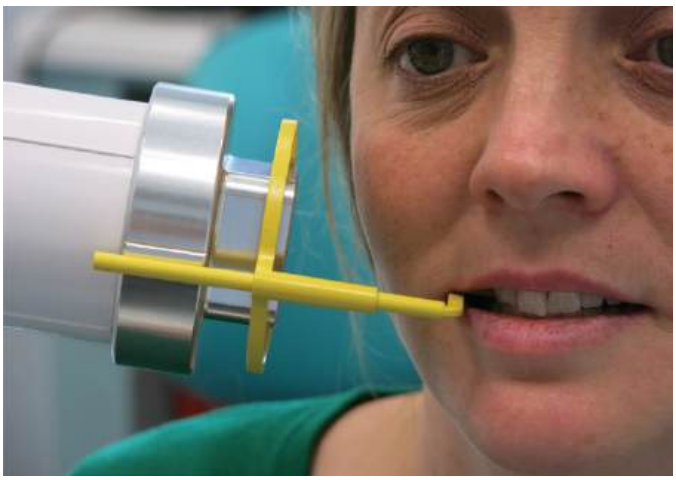


Figure 2.13 Right bitewing radiograph being taken with film holder and beam-aiming device (yellow arm) enabling optimal alignment of the radiographic beam at right angles to the teeth and intra-oral film.

Incipient occlusal lesions are difficult to detect, and only later-stage occlusal lesions are clearly visible on these radiographs. Periapical radiographs may be used to assess the depth of proximal lesions in anterior teeth. Dental panoramic tomogram radiographs (sometimes also called DPTs, OPTs, or OPGs) should not be used routinely for caries detection, due to their limited resolution, potential for distortion, and increased radiation dose by comparison with small intra-oral films. The radiographic appearance of caries (radiolucency in enamel or dentine) is often described as being up to 6 months behind the actual histological spread of the lesion in the tooth. Therefore a lesion radiographically encased within enamel may have actually spread histologically across the enamel–dentine junction (EDJ) into the outer third of dentine. This is because intra-oral radiography, at the clinical radiation dose used conventionally, has only sufficient sensitivity to detect relatively gross changes in mineral density. As the beam passes through quantities of potentially sound, highly mineralized tissues in a tooth, zones of early demineralization will be masked and therefore undetectable on the film/plate/detector. Thus conventional radiography can detect only the demineralization in more extensive enamel lesions and the caries-infected dentine of the deeper lesion (see Chapter 1, Section 1.2.6). Remember that radiographs are two-dimensional representations of the change in radio-density of a three-dimensional object, taken as a snapshot in time. Studies have shown that once a carious lesion has progressed radiographically just into the middle third of dentine, the enamel is likely to be grossly demineralized and cavitated.

Table 2.4 shows how the mICDAS scores for the visual and histological lesion appearance correlate with the radiographic appearance/depth of enamel and dentine lesions. Examples of lesions detected from bitewing radiographs of a high caries risk patient are shown in Figure 2.14 (a) and (b). Once experience has been gained in interpreting radiographs, other clues can often be found to help with the diagnosis and potential activity status of the lesion(s). These might include the outline of the pulp horns—often tertiary dentine has been laid down in response to the active disease process, and this can be observed by the relative shrinkage of the radiolucent pulp horn subjacent to the spreading lesion. In addition, the 'moth-eaten' appearance of the advancing edge of the radiolucency is a clue that, at the time when the film was taken, the lesion was in a state of relative activity causing demineralization. If the dentine–pulp complex has had a chance to lay down extra mineral to 'wall off' the lesion (a more defined radio-opaque boundary), this implies a relative tipping of the metabolic balance towards inactivity and healing.

Table 2.4 The association between the radiographic lesion depth (E1–D3) and equivalent mICDAS scores

Enamel Lesion		mICDAS
E1	Outer half of enamel	0, 1
E2	Inner half of enamel	1
Dentine Lesion		**mICDAS**
D1	Outer third of dentine	2
D2	Middle third of dentine	3
D3	Inner third of dentine	4

It is imperative that the dentist does not immediately start to treat operatively a radiolucency on a radiograph alone, without considering all of the other clinical findings, signs, and symptoms. A diagnosis must be made on the basis of all the information that has been discussed in this section, prior to any operative intervention. Radiographic radiolucencies/radio-opacities in teeth may be caused by:

- pathology (e.g. caries, internal resorption, pulp calcifications)
- natural anatomy (e.g. pulp chamber morphology, rotations/malpositioning of teeth, anatomical superimposition, canine fossae in first premolars)
- artefact (e.g. cervical burnout in the proximal region of the tooth close to the alveolar crest mimicking proximal caries; see Figure 2.14 c)
- restorative process (e.g. radiolucent/opaque restorative materials; see Figure 2.14 c, d, and e).

Figures 2.14c, d, and e show examples of potential conundrums that a clinician may encounter when interpreting the causes of differing radio-densities in bitewing or other radiographs. An 'obvious' initial diagnosis of active caries made from detecting the well-demarcated radiolucency in Figure 2.14c does not correlate with the clinical findings for the same tooth (see Figure 2.14 d). So what is the cause? (See answer to Self-Assessment Question 2.5.) The diffuse radiolucency just subjacent to the restoration in the maxillary molar in Figure 2.14e poses an interesting dilemma. This laminate restoration was placed after minimally invasive caries excavation (see Chapter 5, Section 5.9) and the digital radiograph was taken at the 1-year recall consultation. Such an appearance might be caused by any of the following:

- residual, inactive caries retained during the minimally invasive caries excavation procedure
- active caries due to microleakage at the proximal restoration margin
- the presence of a radiolucent pulp protection material (see Chapter 5, Section 5.11)
- the possible effect on dentine of the restorative adhesive process (e.g. adhesive procedure, dental adhesive; see Figure 7.16f in Chapter 7).

In such cases, it is imperative to examine the tooth/restoration carefully and to check signs and symptoms with the patient. Findings from other special investigations may be required (see later). Collectively, this information will help the dentist to ascertain the cause of the radiographic finding and thus to decide whether it is necessary to manage the situation operatively or preventively. Examination of the patient notes and any previous radiographs and their radiographic reporting will also be useful in this context.

Just taking and filing radiographs in the patient's notes is not acceptable. It is essential that a comprehensive report of all the radiographic findings is documented in the patient's notes for long-term scrutiny, for monitoring, and for dento-legal reasons.

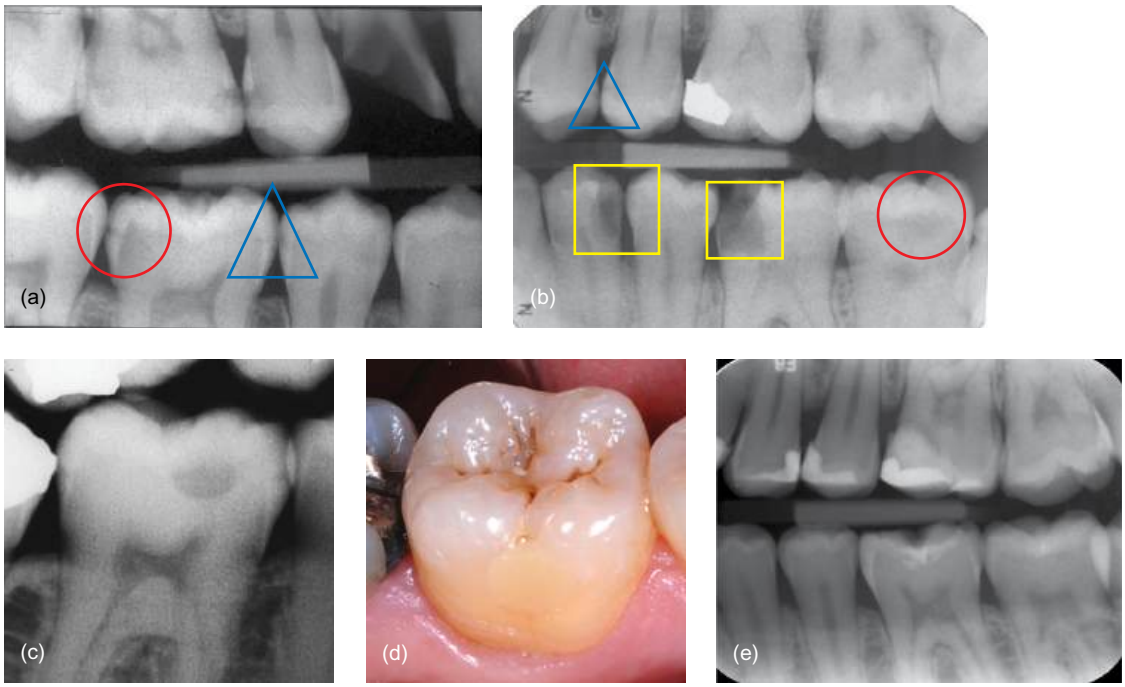


Figure 2.14 **(a)** Right and **(b)** left bitewing radiographs from two different patients, both with high caries experience, showing carious lesions within enamel and outer third dentine (E2–D1, blue triangle), approaching middle third of dentine (D2, red circle), and very close to the pulp (D3, yellow square). Note that other lesions can be detected on these films. **(c)** Well-defined radiolucency in the occlusal aspect of the mandibular molar, as well as radiolucencies present at the cervical aspects both mesially and distally. **(d)** Clinical image of the mandibular molar whose radiograph is shown in (c). **(e)** Left bitewing radiograph focusing on the maxillary and mandibular first molars. Note the radio-opacity associated with the occlusal aspect of the mandibular molar. The maxillary first molar has received a proximo-occlusal restoration to treat a large carious lesion. Note the differences in radio-density between the different restorative materials, enamel and dentine. Note also the diffuse radiolucency between the deepest aspect of the restoration and the pulp chamber (see text for explanation). (Figures 2.14 (c) and (d) courtesy of L Mackenzie.)

Q2.4: Can you find more lesions in Figure 2.14 (a) and (b) and classify them radiographically? Can you comment on the radiographic changes of the pulp chambers in those teeth?

Q2.5: What is the cause of the radiolucencies in Figure 2.14(c)? (Clue—look carefully at (d) for the answer!)

Q2.6:

i Can you guess which materials have been used to restore the cavity in the maxillary molar in Figure 2.14(e)?

ii What other information would you need before deciding to operatively intervene or not, to manage the diffuse radiolucency subjacent to the restoration?

iii What might be the cause of the radio-opacity in the mandibular molar?

Pulp vitality (sensibility) tests

Technically, the term 'vitality' implies the status of pulpal blood flow. This can be measured using laser Doppler flowmetry, which has been mainly applied for research purposes. Clinical signs of a non-vital, necrotic pulp may include the following:

- Discoloration and darkening of the tooth due to the breakdown products of haemoglobin in the pulp chamber. Greying and reduced translucency might also be noticed. These changes may be difficult to detect if the tooth is heavily restored or has an extracoronal restoration covering it.



Figure 2.15 **(a)** A soft, non-tender swelling on the dentoalveolar mucosa distally adjacent to the periapical region of the broken-down UL4, a chronic sinus, into which a sterile gutta-percha point has been carefully inserted **(b)** and radiograph showing the exact location of the 'abscess' **(c)**. The UL6 is vital with no symptoms.

- Over time, a necrotic pulp may give rise to a sinus, tracking from the periapical tissues to the mucosal surface usually adjacent to the apex of the tooth in question. A gutta-percha point inserted gently into the endodontic sinus and then radiographed will show the direction of the sinus track and ultimately the periapical origin of the infection (see Figure 2.15).

The status of pulp innervation (sensibility) can be assessed to ascertain the effect of the caries process on the pulp:

- *Temperature*: heat from warm gutta-percha sticks (rare) or, more commonly, cold from cotton wool pledgets soaked in ethyl chloride or ice sticks may be used to ascertain the status of pulpal innervation. Tooth surfaces to be assessed must be dry and clear of surface debris (plaque/calculus; see Figure 2.16). Check equivalent teeth on the contralateral side to the tooth in question and then the adjacent teeth (acting as an internal control), ensuring that the cotton wool/ice stick/gutta-percha is placed on a clean, dry, sound tooth surface. Ask the patient to raise a hand when they feel a sensation in the tooth being tested. Vital teeth tend to respond quickly, whereas false-positive readings respond more slowly (conduction through dentine/metallic restoration into the periodontal membrane).
- *Electrical*: a monopolar electric pulp test unit passes a small current (direct or alternating) through the patient and the tooth with which it is in contact. The patient's hand must be in contact with the metal handle of the handpiece to complete the circuit with some commercial systems. The probe is placed on a clean, sound tooth surface (contralateral and adjacent equivalents are tested first as an internal patient control) using an electrolytic coupling agent on the tooth surface to ensure completion of the circuit (usually a small amount of prophylaxis paste; see Figure 2.17). The current is increased gradually by the dentist until the patient feels a tingling sensation in the tooth. At this point they are instructed to let go of the handpiece and the circuit is broken. The numerical value can be recorded and is useful for monitoring purposes for a particular tooth, but is not equivocal as the readings can vary from the same patient. False-positive responses may be elicited through stimulation of nerve fibres in the periodontium, and in posterior multi-rooted teeth a mixture of vital and non-vital pulp tissue may confound the interpretation of the reading.
- *Cutting a test cavity without local anaesthesia*: a rarely used, invasive last resort that checks the innervation of the dentine–pulp complex by drilling into potentially vital dentine. If the patient feels pain then at least partial innervation of the pulp remains.

It is important to note in the cases just described that a positive response from the pulp does not necessarily mean all is well. There is no clinical way of knowing if there is partial necrosis or denervation in a dental pulp, and multi-rooted teeth can present with partially diseased



Figure 2.16 **(a)** Pulp sensibility testing of cleaned and air-dried LR4 using cold ethyl chloride-soaked cotton wool pledget. **(b)** Avoid restorations and the gingival margin, as false-positive readings may ensue.

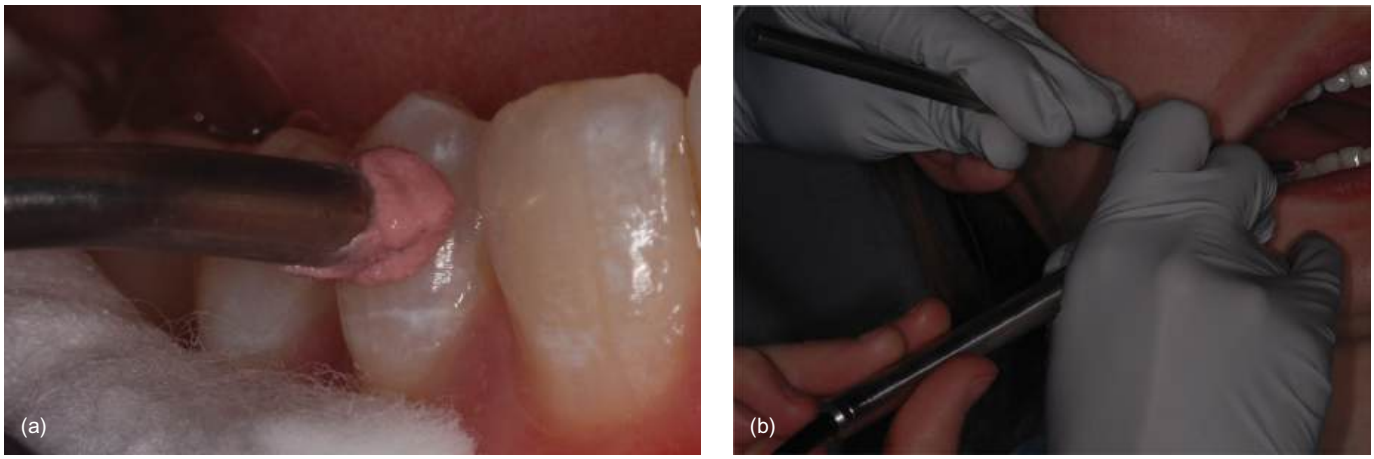


Figure 2.17 **(a)** Probe of the electric pulp tester placed on clean dry tooth surface, coupled with prophylaxis paste. **(b)** Note the patient's fingers (ungloved) touching the probe handle, so completing the electrical circuit. They should let go to break the circuit when a sensation is felt in the tooth.

pulpal tissue. Indeed, the clinical assessment of histological pulp status is not a pure, objective science, and requires interpretation of all clinical findings associated with the tooth in question.

Percussion tests

Percussion tests (gently tapping the crown axially and then obliquely with a probe or mirror handle) assess the physical condition of the periapical tissues and periodontal membrane, but not of the pulp directly. If periapical periodontitis is present, the piston-like effect of the tooth being pushed into the inflamed periapical tissues will elicit acute tenderness. The inflammation in the periodontal tissues might be caused by the toxins from a non-vital pulp.

2.5.4 Lesion activity: risk assessment

Once a carious lesion has been detected, it is important to ascertain whether it is *active* (a state where the disease process is more likely to be progressing or regressing) or *inactive* (a state where it is less likely to be in transition, or is even arrested), so that the correct care plan can be implemented. This is difficult to define and measure objectively intra-orally at present, but certain clues can be gleaned from the following:

- *Colour and surface texture of a cavitated lesion*: an arrested lesion is often darker in colour, and the exposed dentine has a flint-like, hard, shiny outer surface when probed (see Chapter 1, Table 1.1).
- *Presence of gingival bleeding on the same tooth*: the presence of a gingival inflammatory reaction implies poor oral hygiene in that area, which will increase the risk of plaque accumulation, so making it more likely that the lesion will be in an active state of metabolic transition.
- *Presence of plaque*: if a thick biofilm overlies the lesion, it can reasonably be assumed that it is in a state of activity, as it is the presence of the plaque biofilm that is critical to the development of the carious lesion (see Figure 2.18a). If plaque can be found on non-retentive sites, again this implies poor oral hygiene measures and therefore an increased risk of caries incidence.



Figure 2.18 **(a)** Plaque biofilm on the surface of a buccal cervical carious lesion, indicating its potential activity. **(b)** Anterior view of dentition with a 36-hour accumulation of plaque. **(c)** A two-tone plaque indicator applied (dark blue and pink), highlighting the differing densities of plaque accumulated on the labial and gingival margins of the same teeth as in (b).

Q2.7: How could you manage the case shown in Figure 2.18?

- *Accessibility to OH procedures*: if lesions are present in uncleansable sites (see earlier), again it is likely that the lesions will be active, as a biofilm will remain on their surfaces.
- *Use of plaque indicators* (see Figure 2.18b and c): dental manufacturers have produced two- and three-tone plaque-disclosing solutions/gels which can give the practitioner and patient an indication not only of the surfaces on which the plaque biofilm is stagnating, but also for how long, as well as the plaque density and acid levels present at the tooth surface. The bacterial population (and the cariogenicity) of the biofilm develops in a structured manner in correlation with its age—the older the plaque, the more anaerobic and acidic the conditions on the enamel surface are likely to be. These proprietary kits assess neither the direct activity of lesions nor patient susceptibility to caries from the relative quantities, age, and acidogenicity of plaque, but can usefully add to all the other information gathered from the history and examination, so helping to formulate an overall picture of the individual's susceptibility and likelihood to suffer from caries in the future. They can also be helpful as strong motivators for patients, coupled with intra-oral clinical photographs to show them the problems directly, to help to improve their own oral health with better oral hygiene procedures that can be easily demonstrated.
- *Use of saliva tests*: The absence of saliva or the presence of a reduced quality of saliva is an important aetiological factor in the increased susceptibility of patients to caries and tooth wear, and more specifically erosion. More individuals (especially the elderly) are taking multiple medications ('polypharmacy') with the synergistic negative side effect of increased dry mouth (xerostomia), leading to a prevalence of this condition, stated to be affecting approximately 20% of the UK population at present. Volumetric and flow rate analyses can be carried out using simple chairside kits to assess unstimulated and stimulated salivary flow/volume, and crude levels of *Streptococcus mutans* (an indicator of the caries process) can also be calculated. The quality of the saliva can also be documented and monitored using a clinical oral dryness index (see QR code image 2.2 and Table 2.5). If the results from volumetric/flow rate analyses indicate low salivary output and there are high levels of *Streptococcus mutans* in a sample of the patient's saliva, then, *in conjunction with the results of other clinical investigations*, the dentist may collate evidence that points to the patient having a higher caries risk, and any lesions present having an increased likelihood of being active.

QR code image 2.2 Scan this code with your mobile device to view the Challacombe Scale of Oral Dryness, with accompanying images.

www.dentalhealth.org/uploads/download/resourcefiles/download_68_1_The%20Challacombe%20Scale.pdf

Table 2.5 is a text-only version of the scale.

These assessments need to be carried out periodically over time. After initial recorded baseline investigations in patients deemed to

Table 2.5 Challacombe Scale of Oral Dryness (scan QR code image 2.2 for images). This scale provides an additive score ranging from 1 (least severe) to 10 (most severe). Each feature scores 1, and symptoms will not necessarily progress in the order shown, but the total score indicates likely patient needs. Score changes over time can be used to monitor symptom progression or regression.

Score	Clinical findings	Interpretation
1	Mirror sticks to buccal mucosa	An additive score of 1–3 indicates mild dryness. May not need treatment or management. Sugar-free chewing gum for 15 minutes, twice daily, and attention to hydration is needed. Many drugs will cause mild dryness. Routine check-up and monitoring are required
2	Mirror sticks to tongue	
3	Saliva frothy	
4	No saliva pooling in floor of mouth	An additive score of 4–6 indicates moderate dryness. Sugar-free chewing gum or simple sialogogues may be required. Needs to be investigated further if the reasons for dryness are unclear. Saliva substitutes and topical fluoride may be helpful. Monitor at regular intervals, especially for early decay and symptom change
5	Tongue shows generalized shortened papillae (mild depapillation)	
6	Altered gingival architecture (i.e. smooth)	
7	Glossy appearance of oral mucosa, especially palate	An additive score of 7–10 indicates severe dryness. Saliva substitutes and topical fluoride are usually needed. Cause of hyposalivation needs to be ascertained, and Sjögren's syndrome excluded. Refer for investigation and diagnosis. Patient then needs to be monitored for changing symptoms and signs, with possible further specialist input if condition is worsening
8	Tongue lobulated/fissured	
9	Cervical caries (more than two teeth)	
10	Debris on palate or sticking to teeth	

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be at risk of caries, follow-up tests may be performed at predefined, patient-specific recall intervals. This will help to evaluate the patient's risk and susceptibility over time, and thus allow their care to be planned accordingly. Patients categorized as low risk do not necessarily need continuous testing, as this may prove expensive and provide no positive clinical benefit.

2.5.5 Diet analysis

The primary aetiological factors for dental caries, as described in Chapter 1, include a susceptible tooth surface, presence of bacteria, fermentable carbohydrates, and enough time for the combination to start demineralizing the tooth surface beneath the thickened plaque biofilm. The surfaces of the teeth can be examined for carious lesions, and the bacterial levels in plaque and saliva can be investigated (see Section 2.5.4). The other main aetiological factor that can be assessed during this detection phase of patient management is dietary intake of refined carbohydrate and sugar levels in patients exhibiting signs of high caries risk (multiple lesions (> 2) developed within the previous 2 years). This is done by asking the patient to fill out a diet analysis sheet (see Figure 2.19), usually over a period of at least 3 days, encompassing the weekend. They should be advised to write down everything that passes between their lips in that time (including water, the numbers of spoons of sugar in tea/coffee, etc.), and to note when they brush their teeth.

No indication of the relevance of the diet analysis should be given at the first appointment, so as not to bias the patient or allow them to cheat by writing down a diet intake that they feel is an improvement on what they are actually doing! The results should then be analysed by the dentist and/or trained members of the oral healthcare team, and an appointment made to discuss them. Issues to look out for are frequency of meals, hidden sugar content (e.g. many sauces/condiments are high in sugar), medications, and drinks (additional sugar/high sugar content/erosive potential due to their high titratable acidity). It is imperative that the whole team works with the patient to *modify* their existing dietary habits. Chastising the patient and ordering them to change will have the opposite effect! Work with positive reinforcement, offering suggestions or allowing the patient to make sensible suggestions to lower their sugar intake, frequency, and duration. Remember that positive encouragement to instigate small changes over time is most likely to result in positive long-term behavioural adherence. Motivational interviewing skills are a valuable asset in helping to alter patient attitudes and adherence to advice.

	Thursday		Friday		Saturday		Sunday	
	Time	Item ⑧	Time	Item ⑧	Time	Item ⑧	Time	Item ⑧
Before breakfast	7.30	Tea*	7.0	Tea*	7.30	Tea*	7.05	Tea*
Breakfast	8.00	2 Wheat slices 2 Crisp bread 1 Apple Coffee*	8.00	2 Wheat slices 2 Crisp bread 1 Apple Coffee*	8.30	2 Wheat slices 2 Crispbread 1 Apple Coffee*	8.05	2 Wheat slices 2 Crispbread 1 Apple Coffee*
Morning	9.00	Polo	10.00 11.30	Murray mint Tea* Biscuit	11.15	Tea*	10.00 12.30	Lemon Barley Tea*
Mid-day Meal	12.30	Meat roll Tea*	2.00	Steamed fish Parsley sauce Boiled potatoes	1.45	Sausage, onion, Boiled potatoes Ice Cream, tinned fruit	1.40	Roast lamb, potatoes, cabbage, carrots
Afternoon	2.00 5.30	2 Cream crackers 1 Dairy Lea Tea* 2 Shortbread biscuits Tea*	2.45 6.00	Tea* Tea*	2.30 5.45	Tea* Tea*	2.00 4.00	Tea* Tea*
Evening Meal	8.00	Chop, leeks, boiled potatoes Choc-ice Tea*	8.30	Bacon sandwich Tea*	7.30	Fried kipper bread and butter	8.15	Ham salad, bread and butter Tea*
Evening and night	1.00	Horlicks* Biscuits	10.00 1.30	Peanuts Horlicks* Biscuit	9.15 1.45	Chocolate Horlicks* Biscuits	1.15	Horlicks* Biscuits

Figure 2.19 Diet analysis sheet filled in by a middle-aged man with a high incidence of caries.

Q2.8: What stands out as the main problem regarding the sugar intake of the patient in Figure 2.19?

2.5.6 Caries detection technologies

As an adjunct to the principles and methods described previously, there are other technologies available to help the dentist to detect carious lesions. Examples of some of these are shown in Table 2.6. Many detection technologies work on four basic physical principles that affect macro/microscopic changes in mineralized dental tissues, namely optical light scattering, mineral density changes, fluorescent characteristics, and tissue porosity. It is clear from this list that the currently available caries detection devices all work by assessing levels of tissue damage that have already occurred. It is arguable that these technologies do not go far enough—perhaps the ideal detection device is one that detects changes in biochemical/metabolic activity changes which would lead to tissue damage if left unchecked (i.e. detection of lesions before they actually present).

It is important to understand how a particular technology works, in order to be able to interpret accurately the information that it is providing. An example of such an ambiguity is the DiagnoDent laser-fluorescence caries detection system. Note that in Table 2.6, this system has been placed within two categories—optical light scattering and fluorescent characteristics—as clinical research has shown that the unit is affected by both properties to a varying degree. Therefore the clinical interpretation of results—that is, whether to cut a cavity or not—must be undertaken with caution in the clinical environment, again in conjunction with all of the other information gathered for the particular patient. Remember that the best detection technology is a combination of your keen eyes and cerebral processing along with clinical experience gained over time!

2.6 Tooth wear: clinical detection

2.6.1 Targeted verbal history

In patients with tooth-wear lesions, it is vital that not only are these detected and a putative cause attributed (i.e. erosive, attritive, or abrasive lesions from their clinical presentation; see later), but also that the aetiology of the erosion/attrition/abrasion is appreciated by both the patient and the oral healthcare team. In this way, the overall clinical problem can be managed successfully. Coupled to the general information obtained from the initial history as discussed previously, more specific questions may be targeted to assess the aetiology of the specific type of tooth wear, often after the initial clinical examination:

- Relating to erosion:
  - past/present diet (using a diet analysis sheet; see Figure 2.19); questions regarding specific acidic food/drinks using an *aide-mémoire* (see Figure 2.20)
  - digestive disorders causing regurgitation erosion/pregnancy sickness
  - past/present slimming habits, eating disorders, periods of weight loss
  - past/present alcohol intake
  - past/present medications (e.g. vitamin C, iron preparations)
  - past/present occupation—possibility of industrial erosion, rare nowadays.
- Relating to attrition:
  - clenching/grinding habits—day- or night-time
  - levels of stress/anxiety in personal/professional life
  - evidence of masseteric hypertrophy.
- Relating to abrasion:
  - past/present oral hygiene techniques (tooth brushing, flossing, abrasive pastes)
  - habits that cause dental abrasion—pipe smoking, pen chewing, fingernail biting, etc. (see Figure 2.21).

Table 2.6 Examples of modern technologies available to help detect and characterize carious lesions. The four columns indicate the basic physical principle on which each technology relies. The second row gives the technologies useful in the research laboratory

	OPTICAL/ LIGHT SCATTERING	MINERAL DENSITY	FLUORESCENCE	TISSUE POROSITY
In vivo	Modifi ed ICDAS classifi cation Fibreoptic transillumination UV illumination Quantitative laser fl uorescence (QLF) DiagnoDent	Radiography (digital) Radio VisioGraphy Computer tomography (CT) scans (cone beam)	Quantitative laser fluorescence Dye-enhanced laser fluorescence UV illumination DiagnoDent	Electrical conductance measurement (ECM) AC Impedance Dye-enhanced laser fluorescence Dye penetration (iodide) Ultrasound
In vitro	Polarized, transmitted light microscopy (PTLM) Spectroscopy (IR, microRaman) Reflected light confocal microscopy	SEM Quantitative backscattered SEM Microradiography Microfocal CT X-ray microtomography X-ray microanalysis	Confocal laserscanning fluorescence microscopy	Polarized transmission Light microscopy Acoustic microscopy

Name ..... Number ..... Sex M F
Address ..... Tel no .....
..... Occupation .....
..... Work place .....

Interests, hobbies, sport activities .....
Medication .....
Illness present ..... Illness past .....
Has the illness been treated ..... By a doctor ..... In hospital .....

GASTRIC SYMPTOMS	PAST SYMPTOMS			NO	PRESENT SYMPTOMS		
	Frequency per week	Frequency per day	Duration		Frequency per week	Frequency per day	Duration
Belching							
Heartburn							
Acid taste in the mouth							
Vomiting							
Regurgitation ? chew cud							
Stomach ache							
Gastric pain on awakening							
How does the patient treat gastric pain?							
DIET	PAST CONSUMPTION			NO	PRESENT CONSUMPTION		
	Frequency per week	Frequency per day	Duration		Frequency per week	Frequency per day	Duration
Citrus fruits							
Citrus fruit juice							
Other juices							
Special juices							
Sport drinks							
Fruit berries							
Soft drinks, acidic beverages							
Yoghurt							
Vitamin-C drinks, chewable tablets							
Acid sweets							
Special diet							
Vinegar							
Herb tea							
Pickles							
Other acidic food, etc.							
Alcohol							

Figure 2.20 An example of an *aide-mémoire* for use when taking a targeted history for dental erosion.

2.6.2 Clinical presentation of tooth wear

The basic causes of pathological tooth wear (erosion, attrition, abrasion, and abfraction) have been discussed in Chapter 1. The clinical examination outlined in Section 2.4 will permit visual detection of some characteristic features (site/shape of facets/presence of facets on opposing teeth) of the different presentations of tooth wear. However, these are often combined as the cause, and therefore the clinical manifestations, of tooth-wear lesions are multifactorial (see Figure 2.22)



Figure 2.21 An intriguing case of anterior, incisal, undulating abrasion lesions.

Q2.9: What is the cause of the case shown in Figure 2.21?

- *Erosion*: depending on the cause, affects the labial (dietary acids) and palatal smooth surfaces/mandibular buccal/occlusal surfaces (gastric regurgitation acids). Enamel loses its histological surface characterization and becomes smooth and featureless (see Figures 2.23 and 2.24), and eventually grossly dissolves to expose dentine. Can cause smooth-cupped lesions on posterior occlusal surfaces (see Figure 2.24 b).
- *Attrition*: often produces sharp, well-defined, interdigitating tooth-wear lesions on the incisal edges of anterior teeth that meet in occlusion (see Figure 2.23). Can also affect the occlusal surfaces of the molars, but to a lesser extent.
- *Abrasion*: lesions caused by repetitive foreign body contact (e.g. tooth brushing) (classical v-shaped notching on the buccal cervical aspect of mainly anterior teeth).



Figure 2.22 A clinical case of tooth wear showing the multifactorial aetiology, including incisal attrition, labial erosion, and buccal cervical abrasion in a middle-aged patient.

Q2.10: What would be the obvious cause of the buccal cervical abrasion shown in Figure 2.22?



Figure 2.23 A clinical anterior view of a patient with labial–cervical erosion. Note the reflective, smooth, featureless labial enamel surfaces of UR2 and UL2. There are early signs of incisal attrition and the resulting blue-grey translucency.

Q2.11: What might be the presenting complaint of the patient shown in Figure 2.23?

2.6.3 Summary of the clinical manifestations of tooth wear

- The clinical features of faceting/loss of mineralized tissues highlighted in the previous section, cuspal wear (see Figure 2.24b).
- Dental sensitivity/pain caused by active tooth wear as dentine tubules are freshly exposed and pain is elicited on hot and cold stimulation (physiological principles of Brannström's hydrodynamic theory). This is surprisingly uncommon, as tertiary/reparative dentine deposition will often prevent pulp symptoms from occurring.
- Weakened and chipped incisal edges causing trauma to and/or pain from the anterior tip of the tongue (due to roughness), and poor aesthetics (see Figures 2.22 and 2.23).
- Blue-grey appearance of the thinned enamel incisal edges, and possible darkening of teeth due to the change in the optical qualities of the remaining mineralized tissues, again affecting overall aesthetics (see Figure 2.23).
- Loss of tooth structure/fractured teeth or restorations possibly leading to difficulties in chewing due to changes in the occlusion (loss of occlusal contacts, changes in vertical dimension of the bite) and aesthetics (see Figure 2.22).
- Increased risk of marginal fracture/defects of existing restorations as the surrounding hard tissues wear away more rapidly, leaving restorations standing proud.
- Pulpitis, exposure, or even loss of vitality attributable to the extent of tissue loss.

Some of these features may need operative management to provide immediate relief of acute symptoms, or to restore overall dental form and function. However, unless the aetiological factor(s) responsible for tooth wear are discovered, it will never be cured for that particular

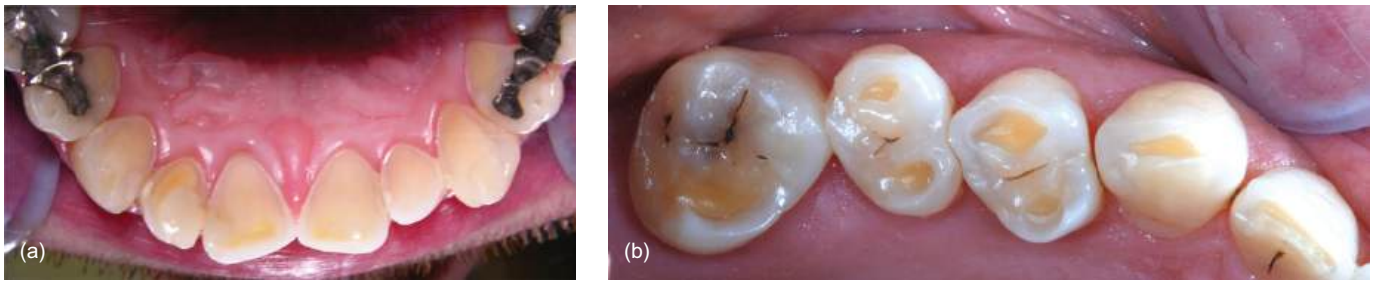


Figure 2.24 **(a)** A mirrored clinical palatal view of an anterior dentition with erosion caused by excessive consumption of dietary acids. Note the smooth surfaces and peripheral rims of enamel on UL3 to UR3, as well as the amalgam restorations in the premolars standing proud of the tooth surfaces. There was no evidence of gastro-oesophageal reflux from the history. **(b)** A clinical image of the maxillary right quadrant of a patient, showing the cupping lesions associated with dental erosion (with an attritive component as well). ((b) Courtesy of L. Mackenzie.)

patient, and the problems will return (see later). Tooth wear also needs careful reviewing over a prolonged period in order to allow assessment of its relative progress (see Chapter 9). As with caries and periodontal disease, several screening indices exist to enable practitioners to ascertain the level and state of progress of the tooth wear, and the prognosis of any treatment provided, and to undertake medium- to long-term monitoring with careful documentation. One such simple screening tool, the Basic Erosive Wear Examination (BEWE), can be used to stage tooth wear in a similar way to that in which the Basic Periodontal Examination (BPE) assesses periodontal disease (see Tables 2.7, 2.8, and 2.9, or scan QR code image 2.3).

www.ncbi.nlm.nih.gov/pmc/articles/PMC2238785

QR code image 2.3 Scan this code with your mobile device to view the Basic Erosive Wear Examination tool.

Table 2.7 Clinical sequence to be followed when using the BEWE

1	Diagnose the presence of tooth wear; eliminate teeth with trauma and developmental defects from the score
2	Examine all teeth and all surfaces of teeth in the mouth for tooth wear
3	Identify in each quadrant the most severely affected tooth with wear
4	Calculate the BEWE score

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Table 2.8 Criteria for grading erosive wear

Score	Features
0	No erosive wear
1	Initial loss of surface texture
2	Distinct defect, hard tissue loss < 50% of the surface area
3	Hard tissue loss ≥ 50% of the surface area

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Table 2.9 Complexity levels as a guide to clinical management

Complexity level	Cumulative score of all sextants	Management
0	≤ 2	Routine maintenance and observation Repeat at 3-year intervals
1	3–8	Oral hygiene and dietary assessment and advice, routine maintenance and observation Repeat at 2-year intervals

(*continued*)

Table 2.9 (*continued*)

2	9–13	Oral hygiene and dietary assessment and advice, identification of the main aetiological factor(s) responsible for tissue loss, and development of strategies to eliminate respective impacts Consider fluoridation measures or other strategies to increase the resistance of tooth surfaces Ideally, avoid the placement of restorations and monitor erosive wear with study casts, photographs, or silicone impressions Repeat at 6- to 12-month intervals
3	≥ 14	Oral hygiene and dietary assessment and advice, identification of the main aetiological factor(s) responsible for tissue loss, and development of strategies to eliminate respective impacts Consider fluoridation measures or other strategies to increase resistance of tooth surfaces Ideally, avoid restorations and monitor tooth wear with study casts, photographs, or silicone impressions Especially in cases of severe progression, consider special care that may involve restorations Repeat at 6- to 12-month intervals

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2.7 Dental trauma: clinical detection

For patients presenting to general dental practice having sustained facial/oral/dental trauma, it is imperative that before any clinical examination is performed, a thorough history is taken of the incident leading to and causing the trauma. Any suspicions of a head injury including facial fractures (from the history, patient behaviour, or general physical examination) should be dealt with by immediate referral to a hospital accident and emergency unit for medical investigation.

Other dentally relevant points to consider are the following:

- *Patient's age*: stage of tooth/root and soft tissue development, and patient compliance issues.
- *Direction of impact/blow/injury*: relevant to the possible direction of displacement of tissues/restorations.
- *Extra-oral soft tissue injuries*: examine cheeks and lip lacerations before investigating intra-orally. Make clear records with diagrams/photographs (after obtaining appropriate consent).
- *Teeth*: can the patient occlude their teeth? Have any teeth been avulsed from their sockets, and if so, for how long and has the patient managed to retrieve the tooth? Tooth displacement (including intrusion), mobility (of the individual tooth or dentoalveolar complex), bleeding from intra-oral soft tissue wounds. Type of dental fracture sustained (see Figure 2.25).

Note that if the interval between the injury and presentation is short, pulp vitality testing and percussion tests will have no diagnostic value, as the teeth are more than likely to have been concussed and the periodontal ligament bruised. Patient compliance during these investigations (including intra-oral radiography) may also be limited! Depending on the severity and symptoms of the injury, it might be worth cleaning up the soft tissue wounds, removing blood clots, performing debridement under local anaesthesia where necessary, and prescribing relevant and necessary antibiotics and pain relief medication at the first visit, and then inviting the patient to return after 2 to 3 days in order to investigate the dental injuries further and initiate a suitably individualized care plan (see later).

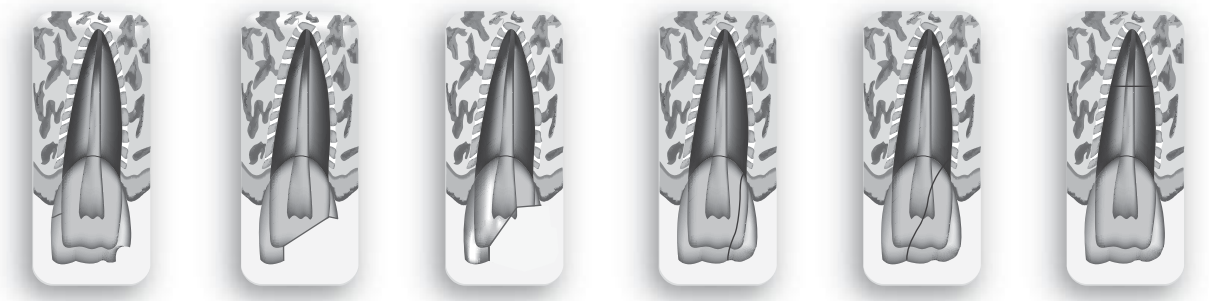


Figure 2.25 Classification of traumatic dental tooth fractures. (Reproduced from J D Andreasen et al. *Textbook and Colour Atlas of Traumatic Injuries to Teeth*, Fourth Edition. © 2007, with permission from John Wiley & Sons Ltd.)

2.8 Developmental defects

As teeth develop and form they are at risk of developing defects that manifest clinically after tooth eruption, often affecting tooth size/shape/number or the quality/quantity of the mineralized tissues themselves. These affected teeth can pose an aesthetic problem, and may also be more prone to the ravages of tooth wear and/or carious attack. These defects have to be detected and diagnosed from the intra-oral dental examination and careful history taking, especially of childhood or maternal illnesses or trauma to the deciduous dentition. Although developmental defects are relatively rare, the aetiology and clinical appearance of some of the more commonly occurring ones have been described in Table 2.10.

Table 2.10 Developmental defects categorized as acquired and hereditary, with descriptions of their basic aetiology and clinical appearance

Developmental defect		Aetiology	Clinical appearance
Acquired	Enamel hypoplasia (see Figure 2.26)	Ameloblast damage (systemic childhood infectious diseases, trauma/infection to deciduous predecessor)	
		Hypoplastic: ↓ matrix, normal maturation	Pitted, thin enamel of normal hardness
		Hypomineralized: normal matrix, ↓ mineralization	Opaque, chalky-white, ?softened enamel
		Systemic cause	Defined areas, bands on all teeth developing at time of illness
	Molar–incisor hypomineralization (see Figure 2.27)	Systemic illness (high fever, respiratory illness) from 0 to 2 years affecting occlusal surfaces of the first molars and possibly maxillary incisors	
		Hypomineralized	White-yellow or yellow-brown opacities, easily chipped, ↑ sensitivity (exposed dentine, plaque stagnation)
	Intrinsic dental fluorosis (see Figure 2.28)	Excessive fluoride ion intake (water, toothpaste, tablets) poisons ameloblast function during enamel formation/maturation	Chalky-white flecks, confluent blotches, brown discoloration, pitted enamel
	Intrinsic tetracycline stain (see Figure 2.29)	Broad-spectrum antibiotic with affinity for mineralized tissues. If taken by mother during pregnancy, deciduous teeth are affected; if taken at age < 12 years, permanent teeth are affected. Rare nowadays	Dark grey horizontal bands affecting all teeth
Hereditary (genetic, familial history)	Hypodontia (oligodontia)	Some teeth do not develop; associated with microdontia. Third molars, second premolars, and upper lateral incisors most affected	Abnormal crown shape/size
	Amelogenesis imperfecta (see Figure 2.30)	Abnormalities in enamel formation (complex classification)	
		Hypoplastic: defect of matrix formation	Thin enamel, yellowish teeth (dentine showing through). Or granular, pitted, stained thin enamel
		Hypomineralized and hypomature: defective matrix mineralization and maturation	Soft, friable, stained, or chalky-white enamel, frequently lost due to weakness at the enamel–dentine junction (EDJ)
	Dentinogenesis imperfecta	Odontoblast defect affecting dentine matrix formation and mineralization. Rare	Enamel lost rapidly due to defective EDJ. Brown, opalescent dentine colour, prone to fracture/wear. Short roots, bulbous crowns, pulps obliterated



Figure 2.26 Pitted hypoplastic enamel in a patient who experienced a severe childhood illness. Note the pattern of teeth affected—maxillary central incisors and tips of maxillary canines and all mandibular incisors approximately a third of the length from the incisal edges and tips of the mandibular canines too.



Figure 2.27 Example of molar hypomineralization with yellow-brown opacities on the occlusal surface of UR6. The disto-palatal fissure shows marked enamel breakdown and demarcated border opacities. (Courtesy of K. Weerheijm.)



Figure 2.28 Severe dental fluorosis with characteristic generalized enamel pitting and brown discoloration. This patient resided in a Sudanese village with natural fluoride levels in the drinking water exceeding 4000 ppm F. The maxillary central incisors have been polished by a dentist using a fine grit bur, so removing the superficial weakened fluorotic enamel.



Figure 2.29 Severe intrinsic tetracycline staining with classic horizontal banding of the stain on the labial surfaces.



Figure 2.30 Severe hypoplastic amelogenesis imperfecta with hard, thin, and pitted enamel remaining on the teeth. The patient presented with two poor-quality acrylic crowns on the upper central incisors, and buccal cervical caries on the lower incisors.

2.9 Suggested further reading and PubMed keywords

Andreasen JO, Andreasen FM, Andersson L. (eds) (2007) *Textbook and Colour Atlas of Traumatic Injuries to Teeth*, 4th edn. Oxford: Blackwell Munksgaard.
<http://eu.wiley.com/WileyCDA/WileyTitle/productCd-1405129549.html>

www.ncbi.nlm.nih.gov/pubmed/advanced

QR code image 2.4 Try searching the following keywords on PubMed for relevant further reading. You can access PubMed by scanning the QR code image or at the address

Keywords

ICDAS; cariogram; CAMBRA; Previser

QR Code image 2.5 Scan this code with your mobile device to view the Cariogram risk assessment system.

www.mah.se/fakulteter-och-omraden/Odontologiska-fakulteten/Avdelning-och-kansli/Cariologi/Cariogram

QR Code image 2.6 Scan this code with your mobile device to view the PreViser™ risk assessment system.

www.previser.co.uk

2.10 Answers to self-test questions

Q2.1: Can you spot an error in the charting in Figure 2.2?

A: In the LL5, there is no MO charted in 'existing restorations', but one has been included in 'treatment to be done.' Well spotted!

Q2.2: What hand instrument should be used instead and for what purpose?

A: A blunt, round-headed dental explorer/periodontal probe.

Q2.3: What treatment may be required for the tooth shown in Figure 2.8?

A: Initial improvement in oral hygiene to remove the plaque effectively from the occlusal surface. If the patient is not compliant, a disto-occlusal restoration may be required.

Q2.4: Can you find more lesions in Figure 2.14 (a) and (b) and classify them radiographically? Can you comment on the radiographic changes of the pulp chambers in those teeth?

A: Mesial UR6, distal UR5: D1—note how even in these relatively distant lesions, tertiary dentine has partly occluded the pulp chambers in these teeth.

Q2.5: What is the cause of the radiolucencies in Figure 2.14 (c)? (Clue—look carefully at (d) for the answer!)

A: The well-defined occlusal radiolucency is caused by superimposition of the radiolucent resin composite restoration that is evident buccally. The proximal radiolucencies are cervical burnout, due to the narrowing of the tooth at this morphological junction between crown and root.

Q2.6:

i. Can you guess which materials have been used to restore the cavity in the maxillary molar in Figure 2.14 (e)?
ii. What other information would you need before deciding to operatively intervene or not, to manage the diffuse radiolucency subjacent to the restoration?
iii. What might be the cause of the radio-opacity in the mandibular molar?

A:

i. The more radio-opaque overlying material is probably a resin composite. The underlying less radio-opaque material is possibly a GIC or Biodentine (see Chapter 7).
ii. The clinical findings with regard to the integrity of the proximal margin, signs/symptoms from the patient/pulp, pulp tests, presence of caries elsewhere in the mouth, review of previous notes/radiographs.
iii. The presence of a resin composite restoration (preventive resin restoration; see Chapter 8).

Q2.7: How could you manage the case shown in Figure 2.18?

A: Improve oral hygiene and minimally restore the cavity to enable easier biofilm removal.

Q2.8: What stands out as the main problem regarding the sugar intake of the patient in Figure 2.19?

A: The number of episodes per day and their frequency throughout each 24-hour period. This would ensure that the patient's teeth were bathed in plaque with a pH below 5.5 for significant periods, leading to demineralization of the enamel.

Q2.9: What is the cause of the case shown in Figure 2.21?

A: The patient plays with a large, metal tongue stud across the anterior teeth, causing a significant amount of tooth-wear damage.



Q2.10: What would be the obvious cause of the buccal cervical abrasion shown in Figure 2.22?

A: Excessive or prolonged action of horizontal tooth brushing.

Q2.11: What might be the presenting complaint of the patient shown in Figure 2.23?

A: The appearance of the incisal edges, chipping of the front teeth, sharpness to the tongue, grey discoloration.